

Metamorphic Circuit Breaker: A Runtime Reliability Layer for Detecting Transform-Brittleness in Deployed Vision Models

Abstract

Machine learning (ML) vision models behave differently from traditional software when they encounter unfamiliar data. Instead of crashing or showing an error, they usually make a prediction with high confidence—even when that prediction is completely wrong. Since models are normally tested using data that is similar to what they were trained on, these hidden failures often go unnoticed. Furthermore, the confidence score produced by the model is not a reliable indicator of whether the prediction is actually correct, especially for out-of-distribution (OOD) inputs.

Several methods have been proposed to detect these unreliable predictions, such as distance-based methods, ensemble models, and reconstruction-based approaches. However, these methods are usually evaluated only as standalone research techniques and rarely provide an automatic way to respond when a problem is detected in a real-world deployment.

To solve this problem, we propose a Metamorphic Circuit Breaker Framework, a reusable safety layer that can be placed around any vision model. Instead of relying solely on the model’s confidence score, the framework checks whether the model’s prediction remains stable after applying a set of semantically-irrelevant transformations to the input image, such as rotation, brightness changes, cropping, and noise. Because not every transformation preserves meaning for every task, each transformation is first validated on a small labeled sample: transformations found to distort ground truth for a given task (for example, large rotations for digit recognition, or aggressive contrast changes for medical imaging) are excluded, so the transformation set is calibrated per deployment rather than applied blindly. A prediction is flagged as unstable only when a calibrated majority of the transformed variants disagree with the original, rather than on a single disagreement, which limits false alarms caused by individually sensitive transforms.

The framework works in two stages. First, before deployment, it performs offline metamorphic testing to identify hidden weaknesses in the model that traditional accuracy measurements cannot detect. Second, during deployment, it continuously monitors incoming predictions using an online circuit breaker. To keep runtime overhead low, the full transformation check is triggered selectively—on low-confidence predictions or a sampled fraction of traffic—and the transformed variants are evaluated as a single batched inference pass rather than sequential calls. If too many unstable predictions are detected within a rolling window, the circuit breaker switches from a Closed state to an Open state. While open, the system stops trusting the model’s predictions and instead routes them to a fallback option, such as a human reviewer, another model, or a predefined safe response; where no such fallback is available, the system degrades to flagging predictions as low-confidence rather than silently serving them. The system periodically retests whether the model has become reliable again before returning to normal operation.

Because the correct labels for incoming data are not available during deployment, the framework is evaluated using a synthetic out-of-distribution dataset created by intentionally modifying a portion of a standard test set with effects such as blur, weather conditions, and previously unseen object classes. The framework is measured using standard OOD detection metrics, in-

cluding AUROC, False Positive Rate at 95% True Positive Rate (FPR95), and detection latency, alongside the added inference overhead introduced by the transformation checks. The framework is positioned as one layer within a broader reliability strategy: prediction-consistency targets transform-brittleness specifically, and is intended to complement, rather than replace, confidence- and distance-based OOD techniques, since stable predictions are not always correct ones.

By adapting the Circuit Breaker design pattern from software engineering to machine learning reliability, this framework not only detects unreliable predictions but also automatically prevents them from reaching end users, while keeping the additional computational cost and false-alarm rate practically bounded.

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